

Investors Are (Not) Always Right: A Comparison of Profitability-based and Transaction-based Evaluation for Financial Asset Recommendations

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Finding the right investments is hard!

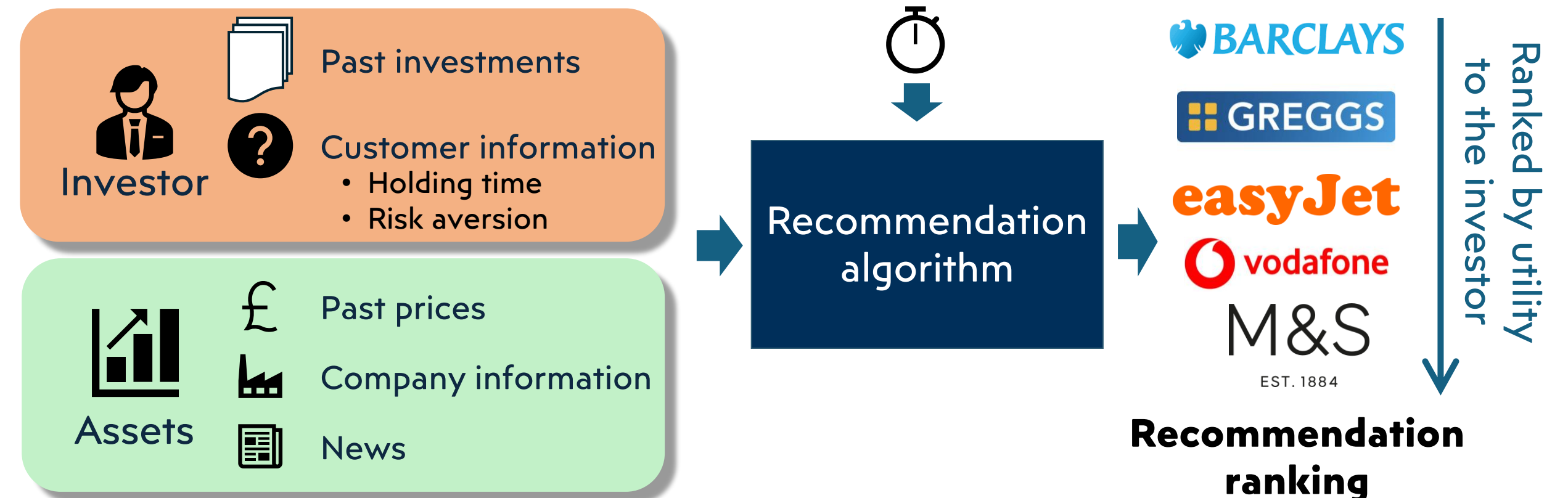
Complexity	Finance is complicated, and many people don't know enough
Time	Investing successfully takes time and effort to research and understand the market
Risk	There are a wide range of investment risks, which are difficult to quantify
Choice	The range of possible investment is huge (thousands of possibilities)
Advice	It is not clear who an investor should trust to get advice

Can we use AI/ML to help people to identify useful investments?

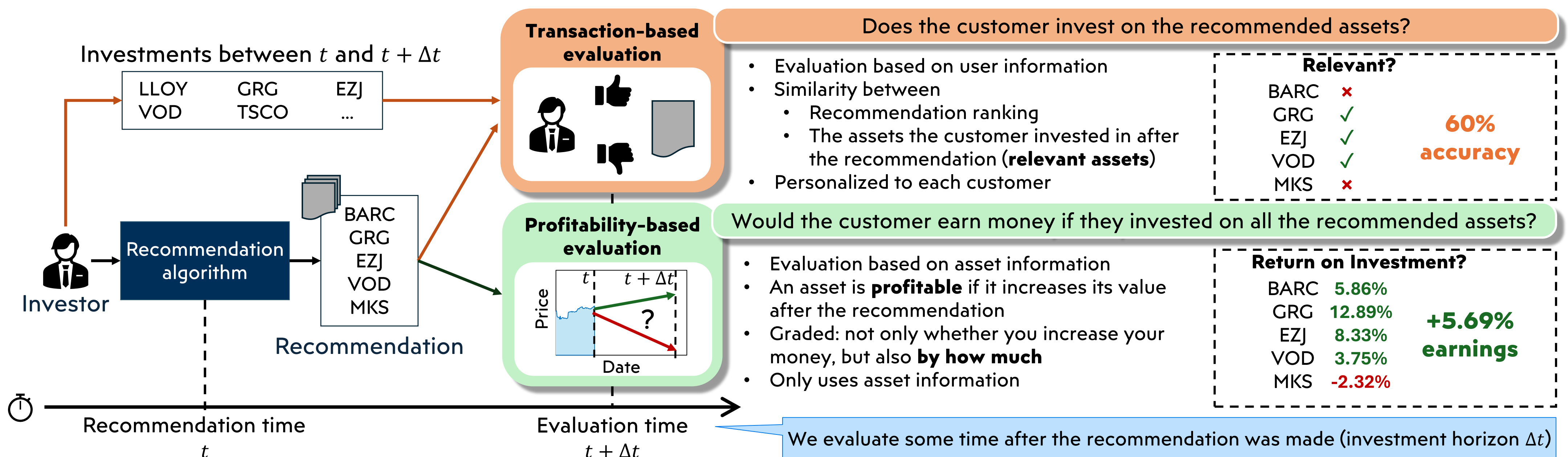


Financial asset recommendations can help!

Goal: Identify suitable investments (stocks, bonds, funds) for a customer



How do we determine whether a financial asset recommendation is good or bad?

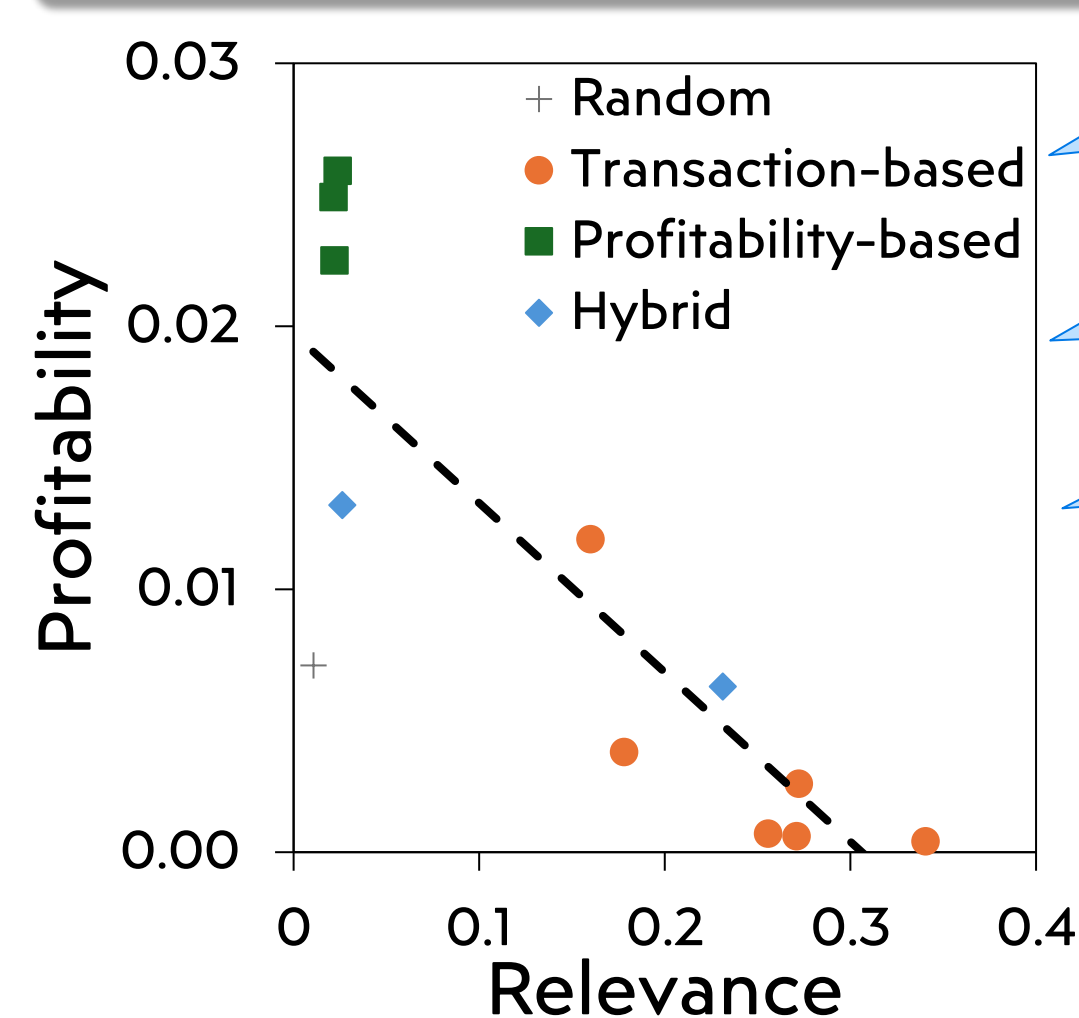


Can we evaluate financial asset recommendations based using ONLY transaction-based evaluation?

Experimental setup

- FAR-Trans dataset** – Snapshot of the Greek Market
 - Period:** 1st Jan 2018 – 30th November 2022
 - Splits:** Every two weeks from 1st Aug 2019 to 23rd Nov 2021
 - Investment horizon Δt :** 6 months
- Algorithms**
 - 6 transaction-based models (using only user information)
 - 3 profitability-based models (using only asset information)
 - 2 hybrid models (using both asset and user information)
 - Random recommendation
- We compare whether two metrics are positively correlated**
 - Transaction-based:** nDCG @ 10
 - Profitability-based:** Return on investment @ 10

Results



- Transaction-based models are good at identifying relevant assets
- Profitability-based models are good at identifying profitable assets
- But they are bad at the other task
- Correlation between metrics is negative!**

NO. If we only rely on transaction-based evaluation, we might make our investors lose money!

Why?

Customers are not necessarily good investors

Their effectiveness varies over time

And their investment strategy also matters

